



FATTY ACID SYNTHESIS

Dr Kwangu, PhD

Fatty acid synthesis

- Important features of this pathway:
 - Synthesis takes place in cytosol; β -oxidation takes place in mitochondrial matrix.
 - Intermediates are bound to sulfhydryl groups of acyl carrier protein (ACP); intermediates of β -oxidation are bonded to CoA
 - Growing fatty acid chain is elongated by sequential addition of two-carbon units derived from acetyl-CoA
 - Reducing power comes from NADPH; oxidants in β -oxidation are NAD^+ and FAD
 - Elongation of fatty acid stops when palmitate (C_{16}) is formed; further elongation and insertion of double bonds carried out later by other enzymes
- Fatty acid synthesis takes place in three stages:
 1. Mitochondrial acetyl-CoA is transported into cytosol via citrate transport system. Acetyl CoA is condensed with oxaloacetate to form citrate ---> antiported out with inward movement of anion. Citrate cleaved by cytosolic ATP citrate lyase --> oxaloacetate + acetyl CoA
 2. Formation of malonyl CoA

Acetyl CoA carboxylase is key regulatory enzyme

Influenced by glucagon --> inactivates enzyme in liver

Epinephrine inactivates enzyme in adipocytes

Citrate allosterically activates enzyme

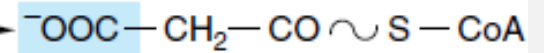
Fatty acyl CoA allosterically inhibits enzyme

Production of malonyl-CoA is the initial and controlling step in fatty acid synthesis

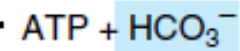
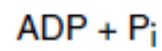
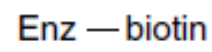
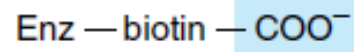
- Bicarbonate is a source of CO_2 in the carboxylation of acetyl-CoA to malonyl-CoA in the presence of ATP and *acetyl-CoA carboxylase*
- *Acetyl-CoA carboxylase* -requires biotin
 - multienzyme protein
- Reaction occurs in two phases
 1. Carboxylation of biotin involving ATP
 2. Transfer of the carboxyl group to acetyl-CoA to form malonyl-CoA



Acetyl-CoA

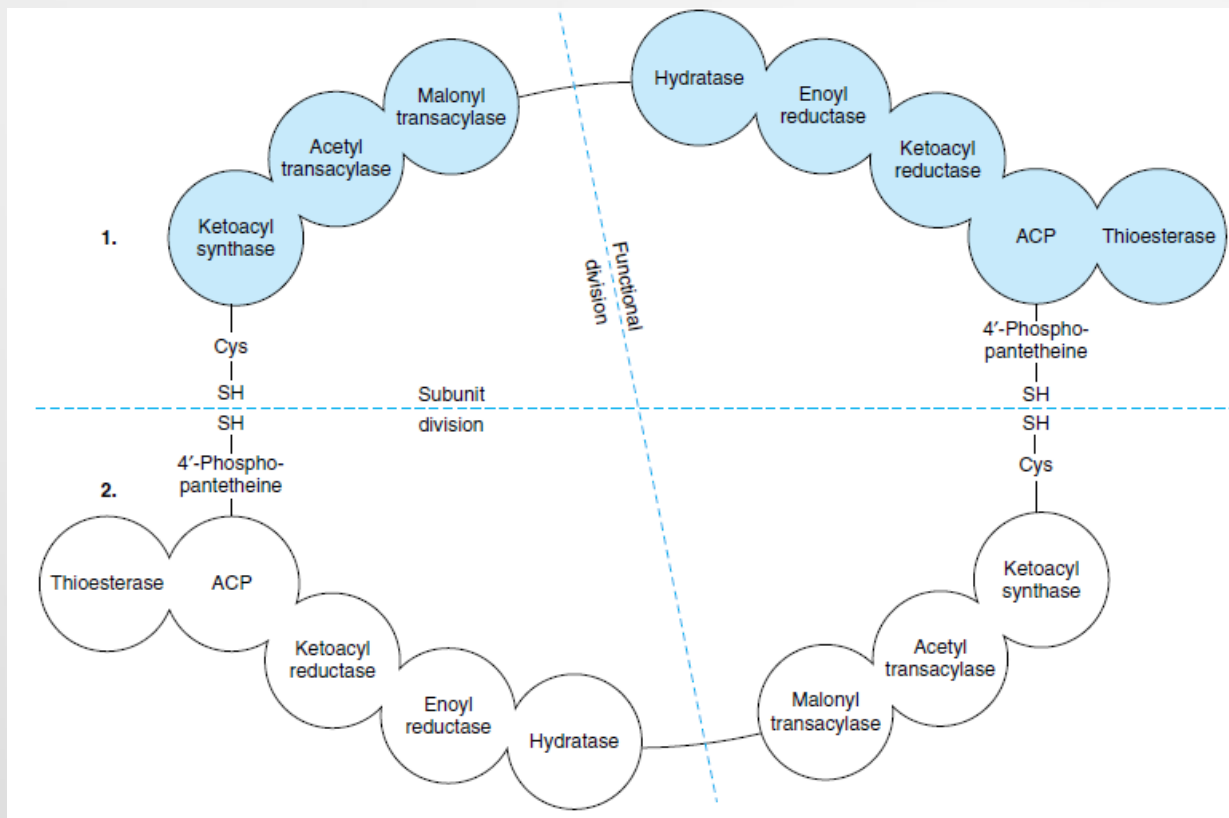


Malonyl-CoA

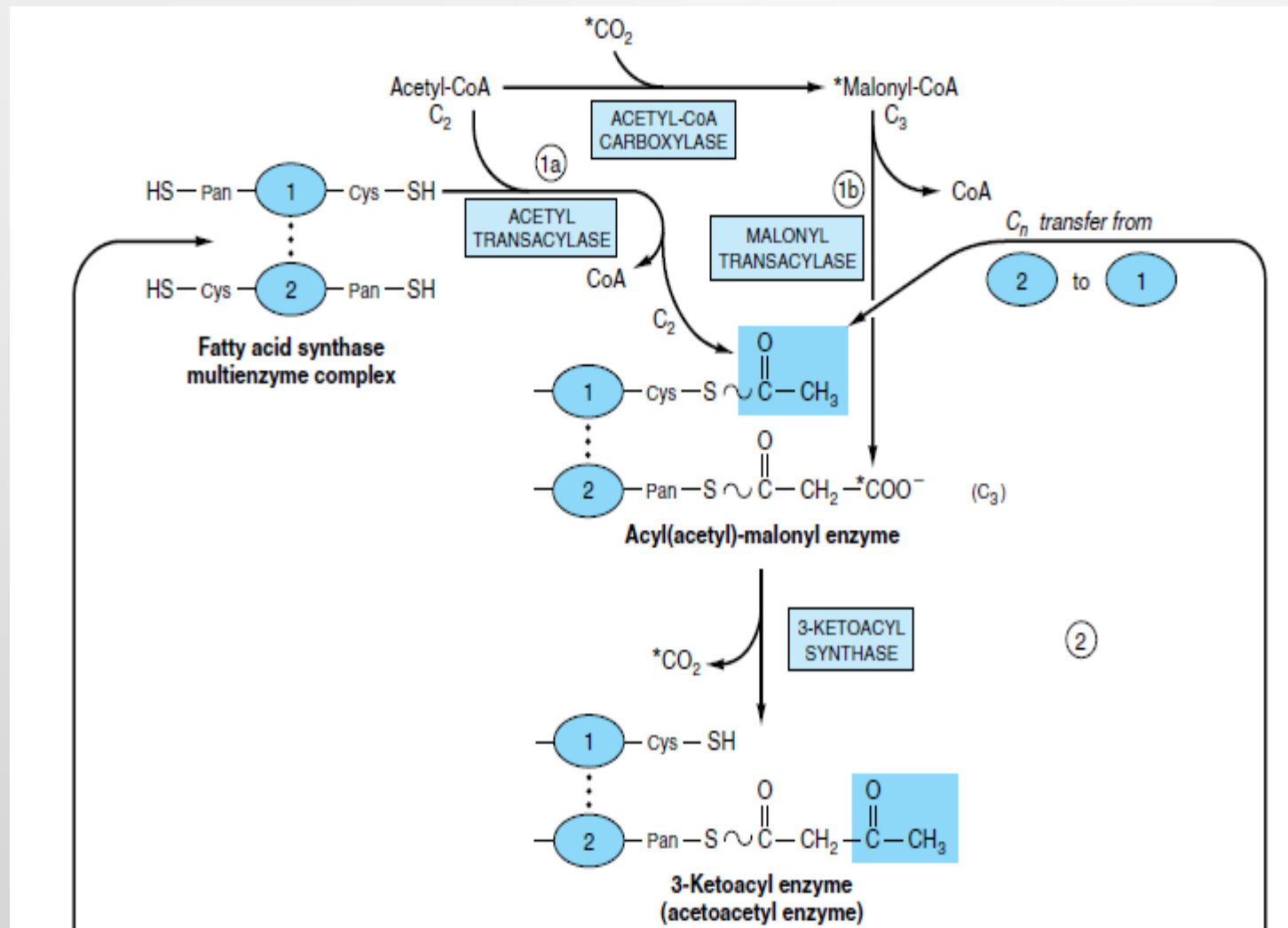


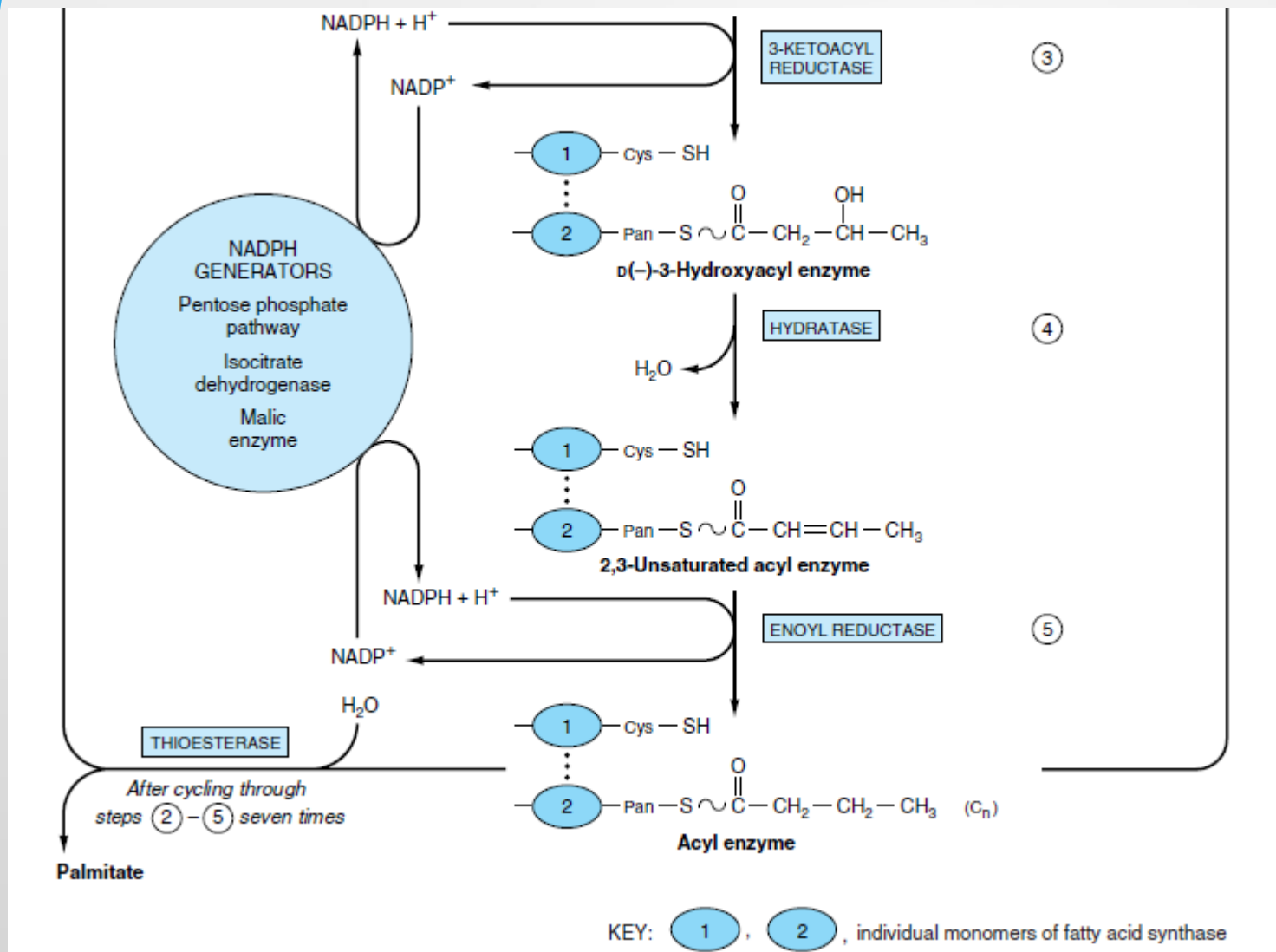
The fatty acid synthase complex is a polypeptide containing seven enzyme activities

- The synthase system is a multienzyme polypeptide complex that incorporates
 - acyl carrier protein (ACP), which takes over the role of CoA
The ACP contains the vitamin pantothenic acid in the form of 4'-phosphopantetheine. Intermediates in fatty acid synthesis are linked to the sulfhydryl terminus of 4'-phosphopantetheine
 - dimer comprising two identical monomers, each containing all seven enzyme activities of fatty acid synthase on one polypeptide chain
- Throughout the process of fatty acid synthesis, the intermediates remain covalently attached to one of the two thiol groups of the complex
 - One point of attachment is the -SH group of the cysteine residue of ketoacyl-synthase
 - The other is the -SH group of acyl carrier protein
- The acyl carrier protein is a small protein containing the prosthetic group 4'-phosphopantetheine
 - Hydrolysis of the thioester that links ACP to the fatty acyl group is highly exergonic and the energy released helps to make the first reaction in fatty acid synthesis (condensation) favourable



Fatty acid synthase multienzyme complex. The complex is a dimer of two identical polypeptide monomers, 1 and 2, each consisting of seven enzyme activities and the acyl carrier protein (ACP). (Cys-SH, cysteine thiol.) The -SH of the 4'-phosphopantetheine of one monomer is near the -SH of the cysteine residue of the ketoacyl synthase of the other monomer, suggesting a “head-to-tail” arrangement of the two monomers. Though each monomer contains all the partial activities of the reaction sequence, the actual functional unit consists of one-half of one monomer interacting with the complementary half of the other. Thus, two acyl chains are produced simultaneously.

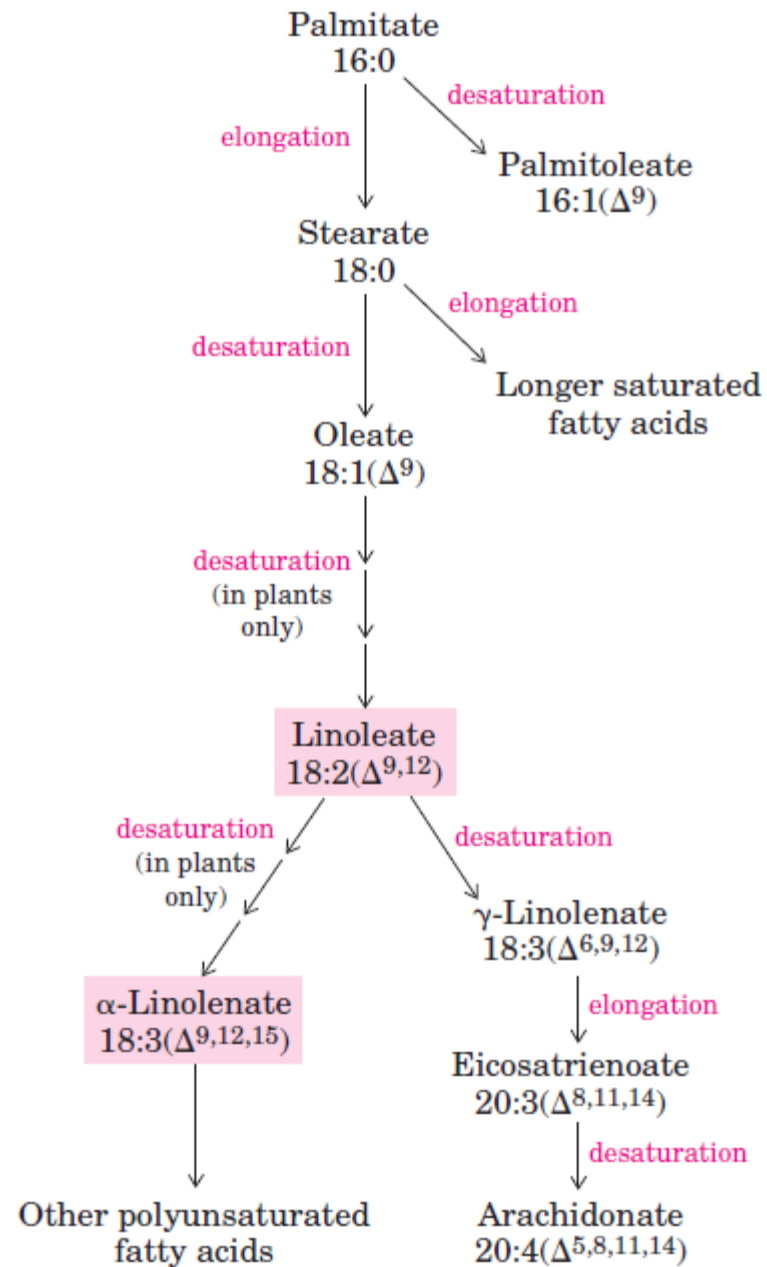




Biosynthesis of long-chain fatty acids. Details of how addition of a malonyl residue causes the acyl chain to grow by two carbon atoms. (Cys, cysteine residue; Pan, 4'-phosphopantetheine.)

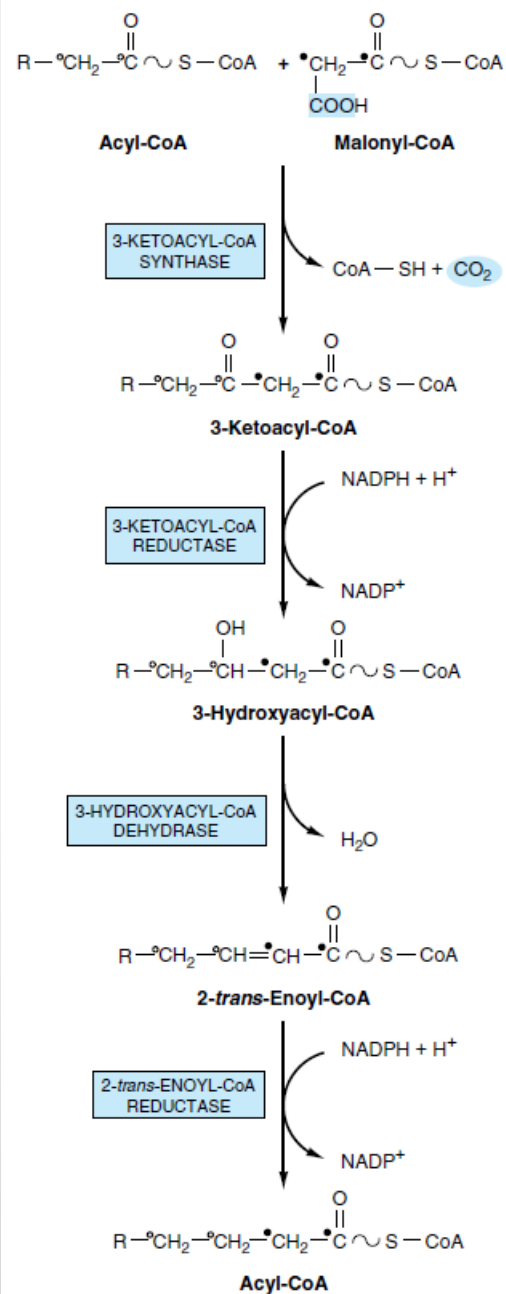
Reaction

- 1a. A priming molecule of acetyl-CoA combines with cysteine –SH group of ketoacyl- synthase catalyzed by *acetyl transacylase*
- 1b. Malonyl-CoA combines with the adjacent –SH on the 4'-phosphopantetheine on ACP of the other monomer catalyzed by *malonyl transacylase* to form acetyl (acyl)– malonyl enzyme
2. The acetyl group attacks the methylene group of the malonyl residue and liberates CO_2 , forming 3-ketoacyl enzyme (acetoacetyl enzyme) freeing cysteine –SH group. Catalyzed by *3-ketoacyl synthase*. Decarboxylation allows the reaction to go to completion
- 3,4 and 5. The 3-ketoacyl group is reduced, dehydrated and reduced again to form to the corresponding saturated acyl-S- enzyme
 - . A new malonyl-CoA molecule combines with –SH on the 4'-phosphopantetheine, displacing the saturated acyl residue onto the free cysteine –SH group.
 - . The sequence is repeated six more times until a saturated 16-carbon acyl (palmityl) has been assembled.
 - . The saturated 16-carbon acyl (palmityl) is released from the enzyme complex by a seventh enzyme *thioesterase*
 - . The free palmitate must be activated to acyl-CoA to proceed in other metabolic pathways
- Acetyl-CoA used as a primer forms carbon atoms 15 and 16
- Addition of all the other C_2 units is via malonyl-CoA
- Propionyl-CoA acts as a primer for synthesis of long-chain fatty acids having odd number of carbon atoms -ruminant fat and milk
- .Fate –esterification into acylglycerols
 - chain elongation or desaturation
 - esterification to cholesteryl ester



Elongation

- Elongation can occur in most tissues; the process differs in the endoplasmic reticulum vs. the mitochondria.
- **in the Endoplasmic Reticulum**
 - In the endoplasmic reticulum, the reactions resemble those of de novo fatty acid synthesis catalyzed by fatty acyl synthase, but the individual activities appear to be on separate enzymes.
 - Notice that malonyl-CoA is the source of the added carbons, as in de novo fatty acid synthesis.
 - The activities are closely associated with the endoplasmic reticulum membranes.
 - No ACP is involved; CoA esters are used directly.
- **In the mitochondrial Minor Pathway**
 - elongation occurs by a reversal of beta-oxidation.
 - The process is essentially a reversal of beta-oxidation, except that one NADPH and one NADH are required (beta-oxidation yields two NADH). Mitochondrial fatty acid elongation acts primarily on fatty acyl-CoA substrates shorter than 16 carbons.

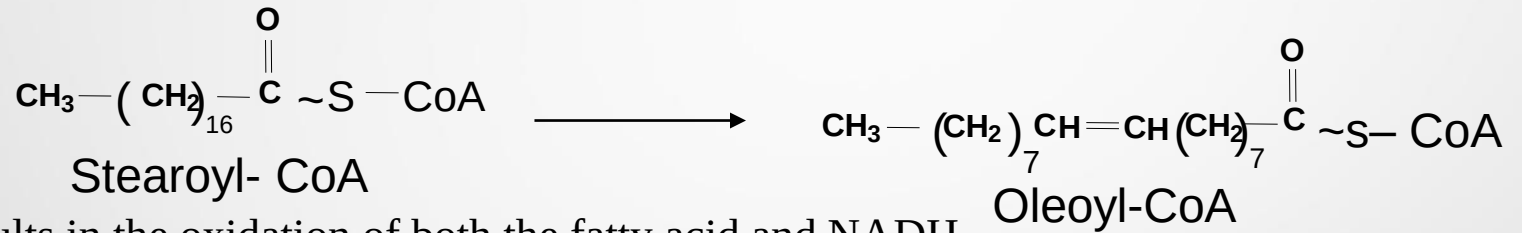


Elongation of Fatty Acid Chains Occurs in the Endoplasmic Reticulum

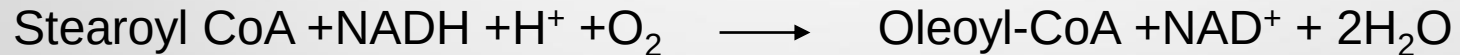
Unsaturated fatty acids

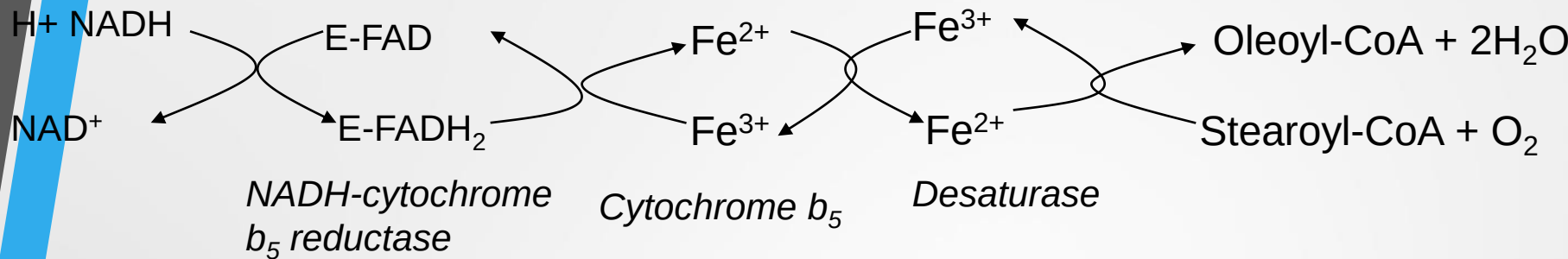
- Unsaturated fatty acids in phospholipids of cell membranes are important in maintaining membrane fluidity
- A high ratio of polyunsaturated fatty acids to saturated fatty acids (P:S ratio) in the diet is a major factor in lowering cholesterol, which is beneficial in preventing coronary heart disease.
- Mammals lack enzymes to introduce double bonds at carbon atoms beyond C9 in the fatty acid chain
 - Hence, can not synthesize linoleic (18:2 cis- Δ^9 , Δ^{12}) and linolenic (18:3 cis- Δ^9 , Δ^{12} , Δ^{15}) – The two essential fatty acids
 - These two essential fatty acids are used to form eicosanoic (C20) fatty acids, which give rise to eicosanoids

molecular oxygen



- Results in the oxidation of both the fatty acid and NADH



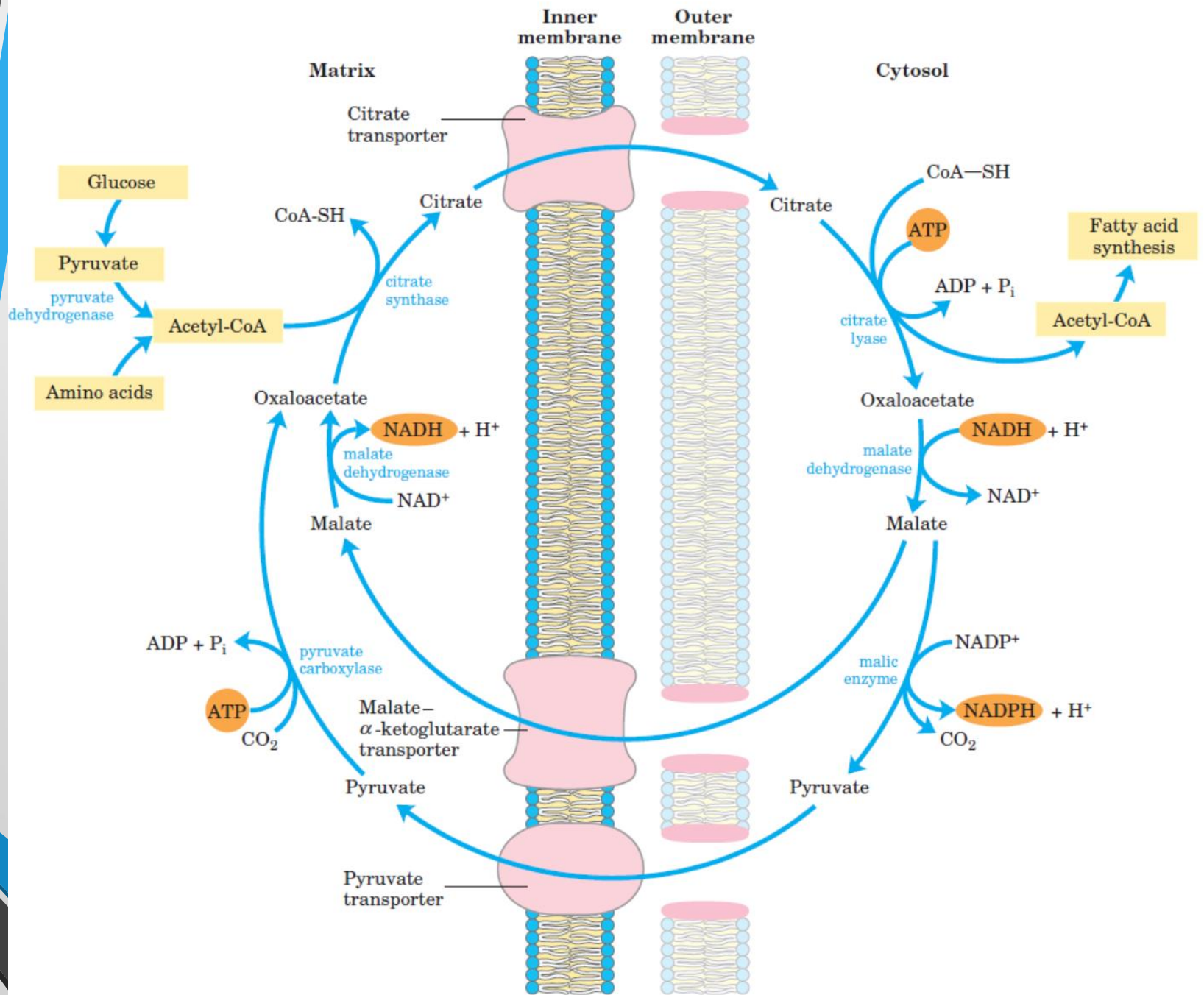


- Three membrane bound enzymes involved; *NADH-cytochrome b_5 reductase*, *Cytochrome b_5* , and *Desaturase*
- Electrons are transferred from NADH to the FAD moiety of *NADH-cytochrome b_5 reductase*
- The heme iron of *Cytochrome b_5* is then converted to the Fe^{2+} state. The nonheme iron atom of *Desaturase* is subsequently converted to *the* Fe^{2+} state, which enables it to interact with O_2 and the saturated acyl-CoA substrate.
- A double bond is formed, and two molecules of H_2O are released.
- A variety of unsaturated fatty acids can be formed from oleate by a combination of elongation and desaturation

- As already mentioned, the human desaturase is unable to introduce double bonds beyond C-9.
- On the other hand, two types of fatty acids - those having double bonds 3 carbons from the methyl end (ω -3 fatty acids) and 6 carbons from the methyl end (ω -6 fatty acids) - are required for the synthesis of eicosanoids (C-20 fatty acids), precursors of important molecules such as prostaglandins, thromboxane and leukotrienes
- Therefore, the ω -3 and ω -6 fatty acids (or their precursors) must be supplied in the diet.
 - As it happens, they are obtained from dietary vegetable oils which contain the ω -6 fatty acid, linoleic acid (C-18:2, $\Delta^{9,12}$) and the ω -3 fatty acid, linolenic acid (C-18:3, $\Delta^{9,12,15}$).
 - Linoleic acid is converted in a series of elongation and desaturation reactions to arachidonic acid (C-20:4, $\Delta^{5,8,11,14}$), the precursor for the synthesis of other eicosanoids in man.
 - Elongation and desaturation of linolenic acid produces eicosapentaenoic acid (EPA; C-20:5, $\Delta^{5,8,11,14,17}$), which is a precursor of yet another series of eicosanoids.

The main source of NADPH for lipogenesis is the pentose phosphate pathway

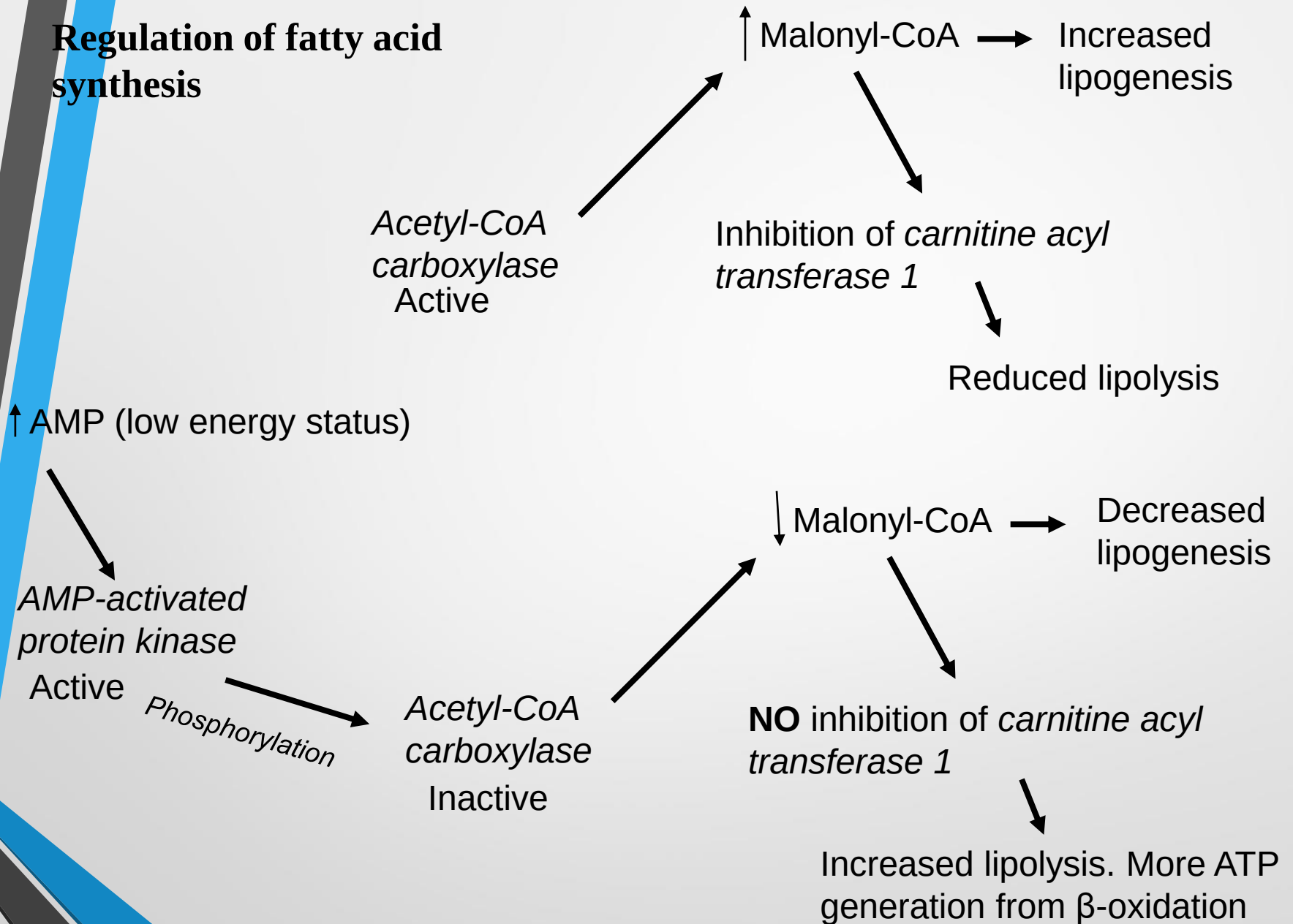
- The oxidative reactions of the PPP are the chief source of hydrogen required for the reductive synthesis of fatty acids
- Tissues specializing in active lipogenesis, i.e, liver, adipose tissue, and lactating mammary gland, also possess an active PPP – both occur in the cytosol, and so there are no barriers against the transfer of NADPH
- Other sources of NADPH – a reaction that converts malate to pyruvate catalyzed by *NADP malate dehydrogenase*“ Malic enzyme” (One NADPH is generated for each acetyl-CoA that is transferred from the mitochondria to the cytosol) - extramitochondrial *isocitrate dehydrogenase*.



- In eukaryotes, nearly all the acetyl-CoA used in fatty acid synthesis is formed in mitochondria from pyruvate oxidation and the catabolism of the carbon skeletons of amino acids
 - Acetyl-CoA from the oxidation of fatty acids does not represent a significant source of acetyl-CoA for fatty acid biosynthesis in animals because the two pathways are reciprocally controlled
- The mitochondrial inner membrane is impermeable to acetyl-CoA, so an indirect shuttle transfers acetyl group equivalents across the inner membrane
- Citrate formed after condensation of acetyl-CoA and oxaloacetate in the CAC is translocated into the cytosol via the tricarboxylate transporter
 - In the presence of ATP and CoA, citrate is cleaved by *ATP-citrate lyase* to acetyl-CoA and oxaloacetate
 - Oxaloacetate can form malate via *NADH-linked malate dehydrogenase*
 - Malate can be converted to pyruvate via *malic enzyme*, with the generation of NADPH
 - NADPH becomes available for lipogenesis, and pyruvate can generate acetyl-CoA after transport into the mitochondria

- In alcoholic individuals, the metabolism of alcohol produces increased amounts of reduced hepatic nicotinamide adenine dinucleotide (NADH). Increased $\text{NADH} + \text{H}^+/\text{NAD}^+$ ratios inhibit the oxidation of fatty acids.
- Fatty acids reaching the liver either from dietary sources or by mobilization from adipose tissue are therefore re-esterified with glycerol to form triglycerides. In the initial stages of alcoholism, these are packaged with apolipoproteins and exported as very-low-density lipoproteins (VLDL).
- Increased concentrations of VLDL, and hence of serum triglycerides, are often present in the early stages of alcoholic liver disease. As the liver disease progresses, there is a failure to produce the apolipoproteins and export the fat as VLDL; accumulation of triglycerides in the liver cells thus ensues.

Regulation of fatty acid synthesis



The nutritional state regulates lipogenesis

- Excess carbohydrate is stored as fat in many animals in anticipation of periods of caloric deficiency. Lipogenesis converts surplus glucose and intermediates such as pyruvate, lactate, and acetyl-CoA to fat
- It's depressed under low restricted caloric intake, on a fat diet, or when insulin is deficient – the latter conditions are associated with increased concentrations of free fatty acids in the plasma
- The nutritional state is the main factor regulating the rate of lipogenesis
- Lipogenesis is increased when sucrose is fed instead of glucose, because fructose bypasses the phosphofructokinase-1 control point in glycolysis and floods the lipogenic pathway
- In summary, rates of fatty acid synthesis are greatest when an individual follows a high-carbohydrate/low-fat diet and are low during fasting/starvation or when eating a high-fat diet. Situations in which there are high circulating concentrations of fatty acids lead to marked inhibition of lipogenesis.

- Short- and long-term mechanisms regulate lipolysis
- Long-chain fatty acid synthesis is controlled in the short term by allosteric and covalent modification of enzymes and in the long term by changes in gene expression governing rates of enzyme synthesis

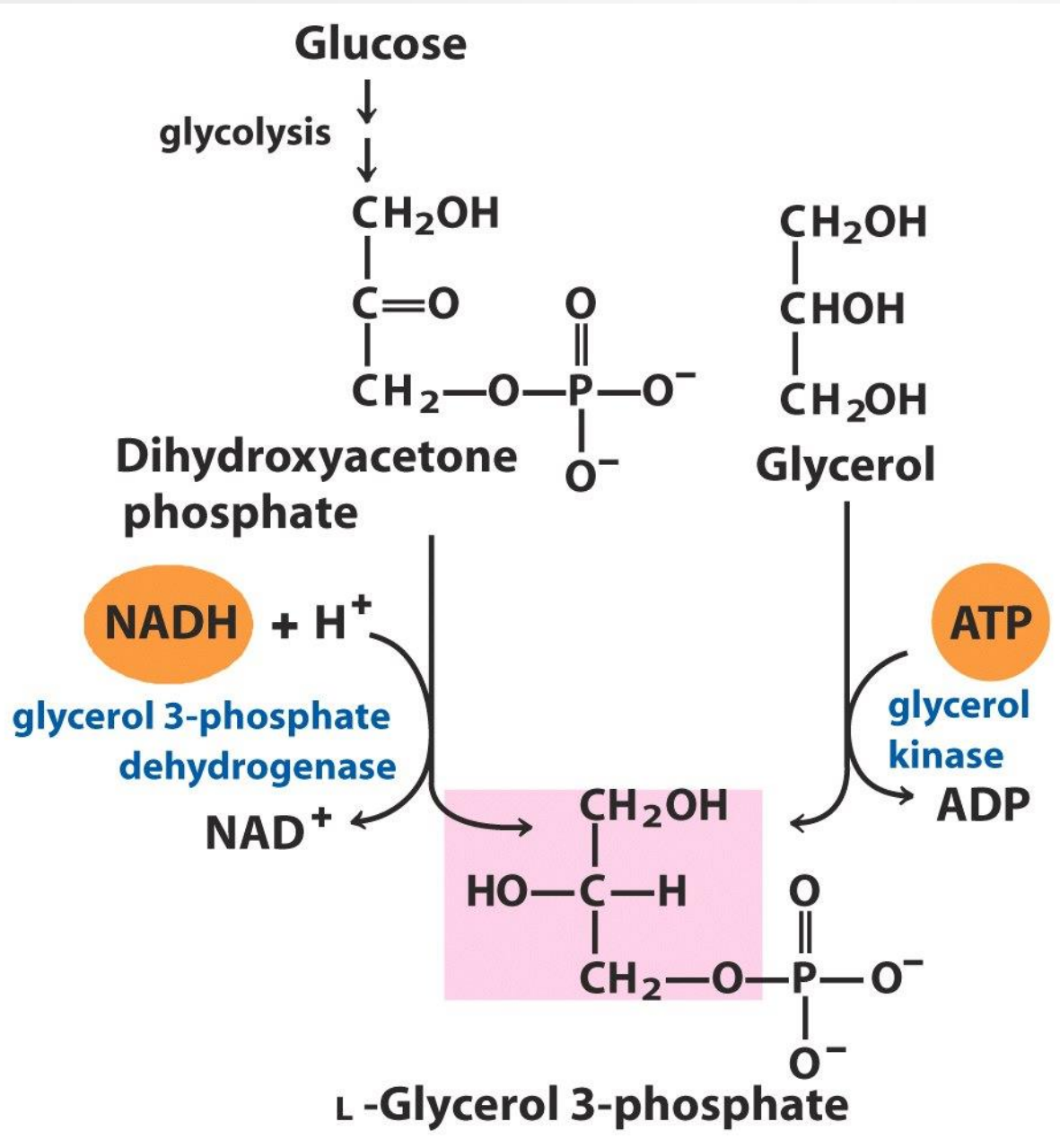
Acetyl-CoA carboxylase is the most important enzyme in the regulation of lipogenesis

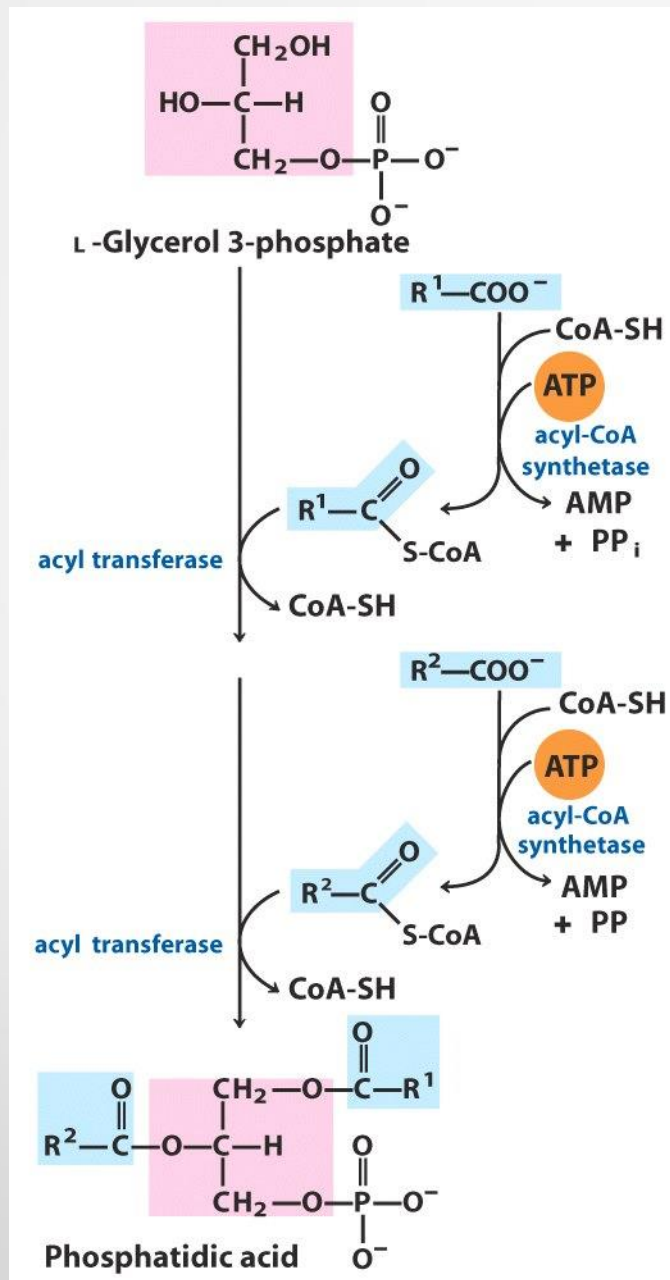
- Activated by
 - citrate
 - Insulin (? Insulin stimulated protein kinase- ?leading to activation of protein phosphatase 2A)
- Inhibited by
 - Phosphorylation of the enzyme
 - Glucagon and epinephrine via cAMP cascade led to phosphorylation and inhibition of acetyl-CoA carboxylase
 - AMP via AMP-activated protein kinase
 - Long-chain acyl-CoA molecules (negative feedback inhibition)

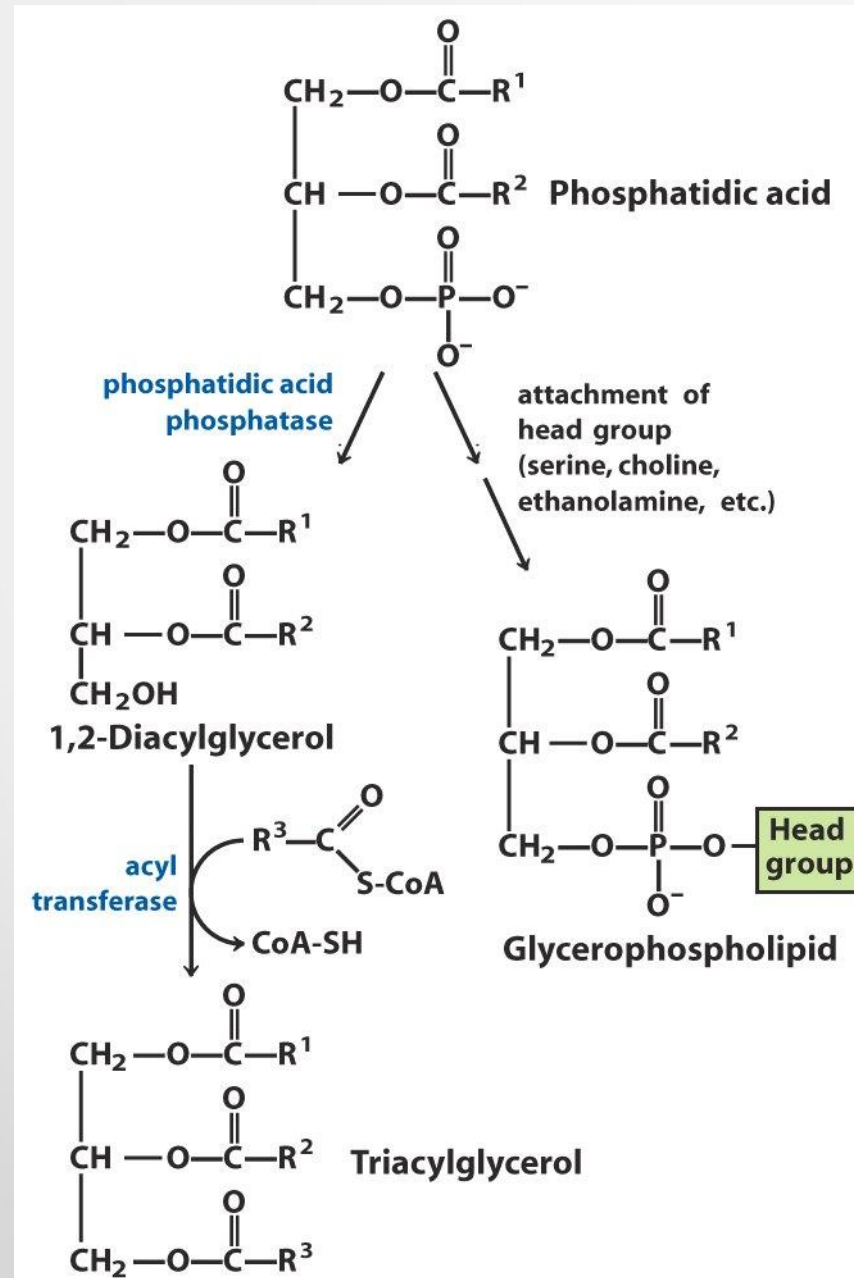
Insulin also regulates lipogenesis/lipolysis by other mechanisms

- . Increases transport of glucose into the cell (adipose tissue), increases the availability of pyruvate for fatty acid synthesis, and glycerol 3-phosphate for esterification
- . Activates pyruvate dehydrogenase in adipose tissue but not in liver
- . Decreasing the level of cAMP **inhibits lipolysis** in adipose tissues and thereby reduces the levels of plasma free fatty acids and, therefore, long-chain acyl-CoA, an inhibitor of lipogenesis.

Biosynthesis of triacylglycerols







- In both the liver and the adipose tissue, triacylglycerols (TAGs) are produced by a pathway involving glycerol-3-phosphate (glycerol-3-P) and phosphatidic acid as intermediates.
- Glycerol-3-P is of different origin in the two tissues: in the liver, glycerol itself provides the source, via the action of glycerol kinase, but in the adipose tissue (which lacks this enzyme), glucose is the source of glycerol-3-P, via glycolysis and its immediate precursor dihydroxyacetone phosphate (DHAP)
- Thus, the storage of fatty acids in the adipose tissue can take place only when glycolysis is activated in the fed state.